

Run Nanopore MinION sequencer till “end of life” of flowcell

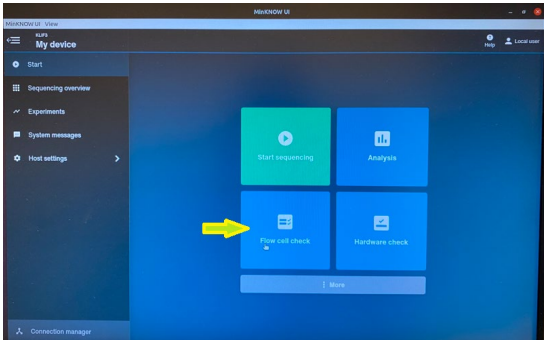
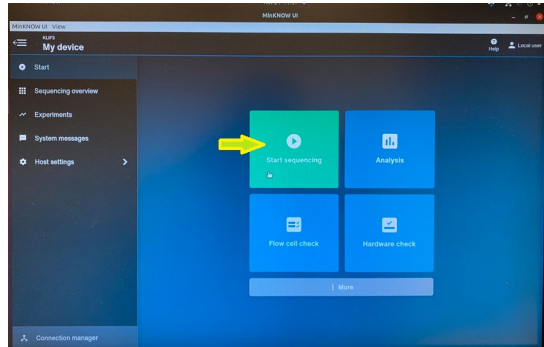
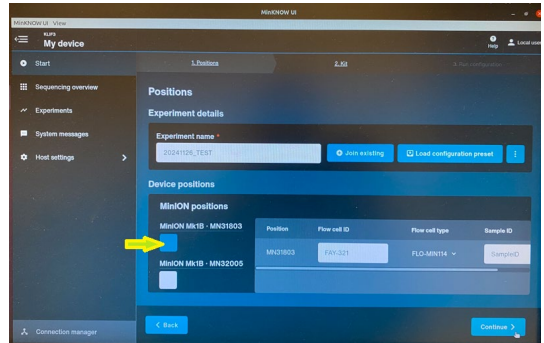
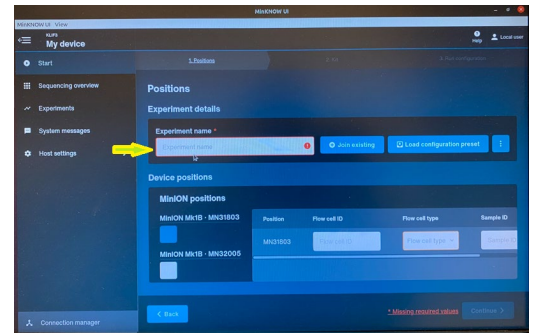
Version: April 2025



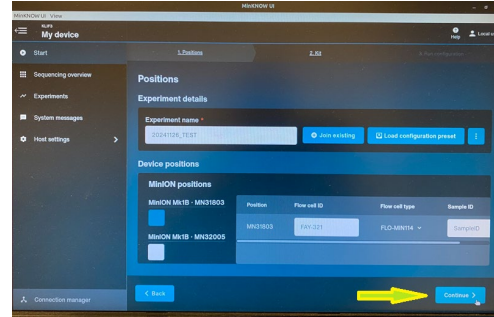
Before Nanopore sequencing:

- Use most recent Nanopore sequencing protocol on Nanopore website
 - Protocol for Rapid Barcoding Kit 96
 - Link for V14: [Nanopore store: Rapid Barcoding Kit 96 V14](#)
- Link how to load a Flow cell: [Priming and loading your flow cell](#)
- Flowcell check: Check Flowcell at arrival, minimal 800 pores
- Check disc space before starting the sequence run (1 TB SSD):
 - If data stored as fast5 and fastQ format: every 1 Gbase of sequence data takes up ~6,5 Gbytes of space
 - If data stored as fastQ only: every 1 Gbase of sequence data takes up ~0,5 gbytes of space
- Useful tips: Nanopore learning (account is necessary): [Getting Started](#)

Manual to run Nanopore MinION sequencer

Insert Flow cell in device Start software Select <Flow cell check> The flowcell should have minimal 800 pores	 The screenshot shows the 'My device' screen in the MinION software. On the left is a sidebar with 'Start', 'Sequencing overview', 'Experiments', 'System messages', and 'Host settings'. The main area contains four large buttons: 'Start sequencing' (green), 'Analysis' (blue), 'Flow cell check' (blue), and 'Hardware check' (blue). A yellow arrow points to the 'Flow cell check' button. At the bottom is a 'More' button and a 'Connection manager' icon.															
Select <Start Sequencing>	 This screenshot is identical to the previous one, but the yellow arrow now points to the 'Start sequencing' button.															
Select MinION device	 The screenshot shows the 'Positions' screen. Under 'Experiment details', there's a text field for 'Experiment name' with 'EXPERIMENT' and buttons for 'Join existing' and 'Load configuration preset'. Below is the 'Device positions' section with a table: <table><tr><th>MinION positions</th><th>Position</th><th>Flow cell ID</th><th>Flow cell type</th><th>Sample ID</th></tr><tr><td>MinION Mk1B - MK31803</td><td>MK31803</td><td>FKV321</td><td>FLO-MIN14</td><td>SampleID</td></tr><tr><td>MinION Mk1B - MK32005</td><td></td><td></td><td></td><td></td></tr></table> A yellow arrow points to the first row of the table. At the bottom are 'Back' and 'Continue' buttons.	MinION positions	Position	Flow cell ID	Flow cell type	Sample ID	MinION Mk1B - MK31803	MK31803	FKV321	FLO-MIN14	SampleID	MinION Mk1B - MK32005				
MinION positions	Position	Flow cell ID	Flow cell type	Sample ID												
MinION Mk1B - MK31803	MK31803	FKV321	FLO-MIN14	SampleID												
MinION Mk1B - MK32005																
Fill in your experiment name	 This screenshot is identical to the previous one, but the 'Experiment name' field now contains the text '12'. A yellow arrow points to this field. At the bottom right, there is a red error message: 'Missing required values'.															

Select <continue>



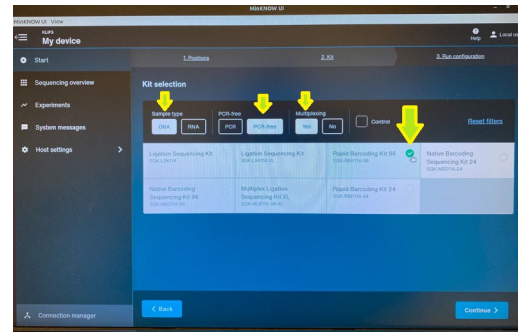
Select <DNA>

Select <PCR-free>

Select Multiplexing <Yes>

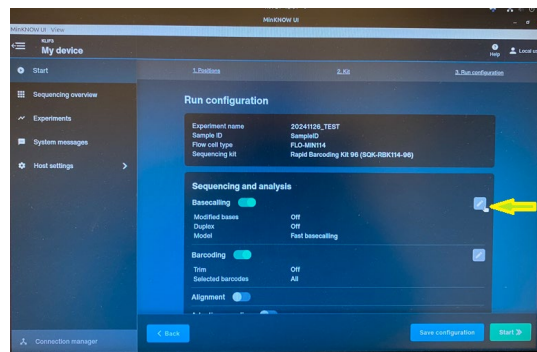
Select <Rapid Barcoding Kit 96
SQK-RBK114-96>

Select <Continue>

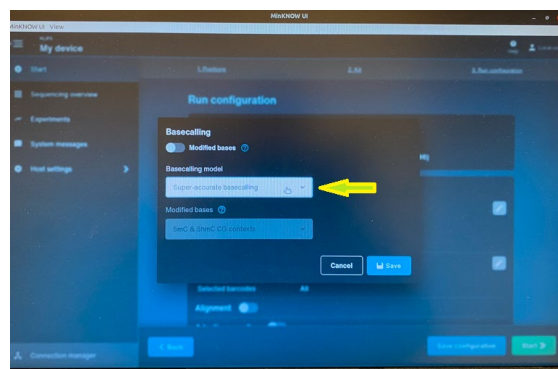


Go to “Run Configuration”

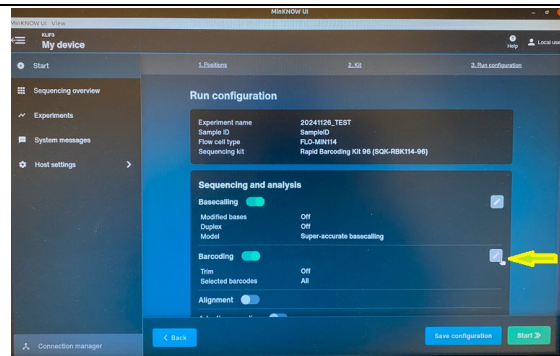
Go to “Sequencing and analysis”



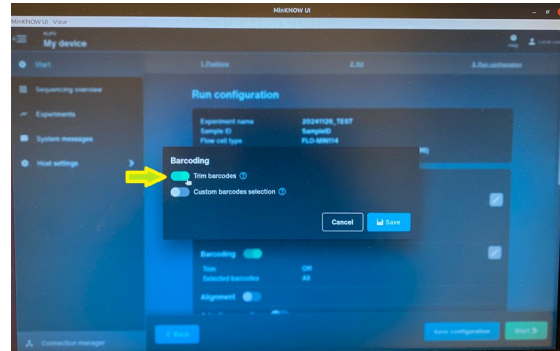
Go to Basecalling Model:
Select <Super Accurate> instead of
Fast basecalling



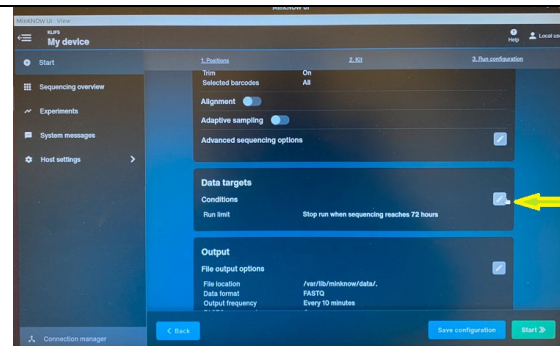
Go to Barcoding in Run configuration



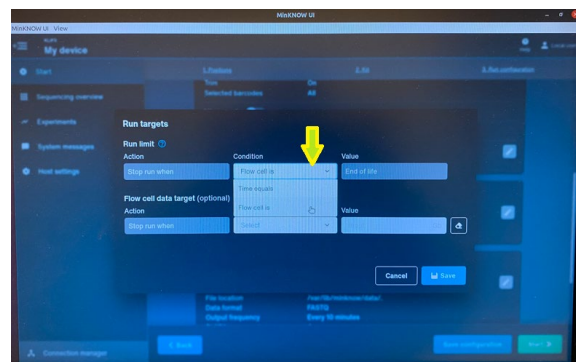
For Barcoding:
Select Trim barcodes <On>
<Save>



Change Run limit/Action
Go to Data targets



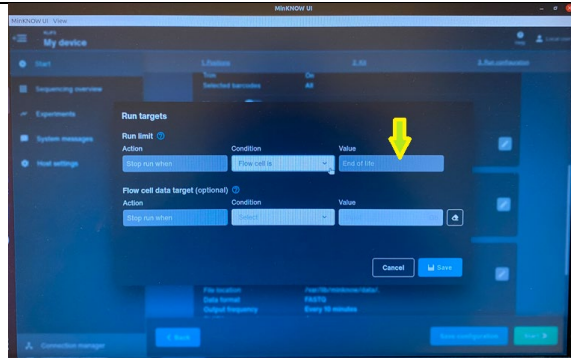
For Data targets:
Select Condition: stop run when....
<Flow cell is>



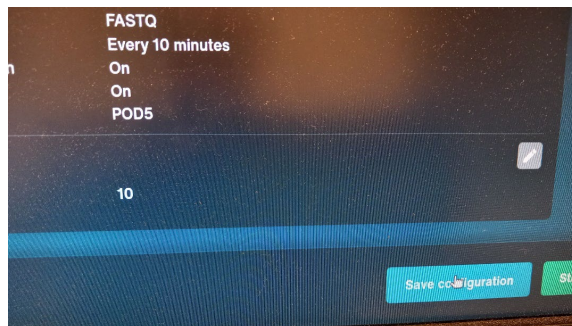
At Run limit:

Select Value: **<End of life>**

<Save>



Save configuration



Select **<Start run>**

Run will stop automatically when Flowcell is fully used.
This will be approx. 3 days.

Flowcell can't be used again after it has run until 'end of life' (it is also no option to wash and use it again).

